

Fereshteh Fathiziarani

fereshtefathiz79@gmail.com | +98 992 464 7100 | Qazvin, Iran | [GitHub](#) | [LinkedIn](#)

EDUCATION

B.Sc. in Psychology

2021 – 2025

University of Danesh Alborz, Qazvin, Iran

- GPA: 19.43 / 20 (3.9 / 4.0) — Ranked 1st in class
- Thesis: Functional Connectivity in Bipolar Disorder (fMRI)

Diploma in Experimental Sciences

2017

GPA: 18.15 / 20

RESEARCH EXPERIENCE

Undergraduate Thesis — Functional Connectivity in Bipolar Disorder

2024 – 2025

Supervisor: Dr. Qazal Taghikhani

Project: Altered Functional Connectivity in Bipolar Disorder

- Investigated task-based functional connectivity in individuals with bipolar disorder and healthy controls using the UCLA CNP open dataset. Focused on the Task-Switching paradigm for the thesis; currently extending analysis to the Balloon Analog Risk Task and Stop Signal Task within a larger collaborative study.
- Designed full preprocessing pipeline in CONN toolbox / SPM12
- Conducted ROI-to-ROI and seed-to-voxel connectivity analyses.
- Contributed to writing (manuscript in preparation).

Computational Neuroscience Research — Neuromatch Academy

2025

Project: Tracking Semantic Representations Across the Visual Hierarchy

- Applied Representational Similarity Analysis (RSA) to fMRI data (Kay et al., 2008) to examine how semantic similarity is encoded across the visual hierarchy (V1–V4, V3A/B, LatOcc).
- Developed a weighted hierarchical semantic similarity matrix; computed Pearson-based neural similarity matrices per ROI and correlated them with semantic structure using Spearman rank correlations.
- Implemented full analysis pipeline in Python; contributed to hypothesis formulation, visualization, and group presentation.

Neuroscience & Cognitive Sciences — Interdisciplinary Schools

2024

- Completed intensive interdisciplinary training in neuroanatomy and neuroimaging; including hands-on assignments in Python and MATLAB.
- Gained practical experience with full fMRI preprocessing pipelines using SPM12 and CONN, including denoising, realignment, normalization, and neuroimaging data interpretation.

PUBLICATIONS & MANUSCRIPTS

Fathiziarani F et al. (in preparation). Altered Functional Connectivity Across Cognitive Tasks in Bipolar Disorder: An Open fMRI Dataset Study.

TEACHING EXPERIENCE

Teaching Assistant — Cognitive Psychology

2025

University of Danesh Alborz | Topics: neuroanatomy, cognition, perception, research design

Teaching Assistant — Application of IT in Psychology

2024

University of Danesh Alborz | Topics: SPSS (t-tests, ANOVA, regression, data cleaning)

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, Matplotlib)

Neuroimaging: CONN toolbox, SPM12, FreeSurfer

Statistical Software: SPSS

Methods: RSA, ROI-to-ROI & seed-to-voxel connectivity, fMRI preprocessing, general linear model (GLM)

TRAINING & WORKSHOPS

Computational Neuroscience — Neuromatch Academy

July 2025

Intensive summer program in computational and systems neuroscience

Neuroscience & Cognitive Sciences — Interdisciplinary Schools

July-December 2024

Neuroanatomy, neuroimaging, and cognitive science methods

Introduction to Neurofeedback

May 2023

Paarand Specialized Center for Human Enhancement

CLINICAL EXPERIENCE

Supervised Neurofeedback Practicum — Fanoos Aramesh Counseling Center July 2023

- Set up and calibrated neurofeedback equipment, guided patients through EEG-based protocols, and monitored real-time brain activity during sessions under clinical supervision.

LANGUAGE PROFICIENCY

IELTS Academic (2025): Overall 7.5 | Listening 8.0 | Reading 8.5 | Writing 6.5 | Speaking 7.0

Persian: Native

RESEARCH INTERESTS

Cognitive Neuroscience · fMRI · Functional Connectivity · Visual Perception · Semantics ·
Psychiatric Neuroimaging · Representational Similarity Analysis · Bipolar Disorder

REFERENCES

Ladan Shahshahani, Ph.D.

Postdoctoral Research Associate, Brown University

ladan_shahshahani@brown.edu

Zahra Ghanbari, Ph.D.

Assistant Professor, University of Danesh Alborz

Zahraghanbari@alborzq.ac.ir

Leila Torabian, Ph.D.

Assistant Professor, University of Danesh Alborz

Leila.torabian@alborzq.ac.ir